

SAI DHIREN MUSALOJI

musalojidhiren@gmail.com | +1 862-423-8830 | [LinkedIn](#)

EDUCATION

- Master of Science in Data Science | *New Jersey Institute of Technology*
- Bachelor of Technology in Electronics & Communications | *Mahatma Gandhi Institute of Technology*

TECHNICAL SKILLS

Languages: Python, Java, C, SQL

AI & LLM Frameworks: LangChain, LangGraph, LlamaIndex, CrewAI, AutoGen, DSPy, Model Context Protocol (MCP)

LLM Providers & APIs: OpenAI, Azure OpenAI, Anthropic Claude, Google Vertex AI, NVIDIA Nemotron, HuggingFace Transformers, Ollama, Mistral AI

Vector Databases & Search: Pinecone, ChromaDB, FAISS, Weaviate, Qdrant, PGVector, Azure AI Search

ML & Data Tools: PyTorch, TensorFlow, Scikit-learn, XGBoost, PySpark, Pandas, NumPy, MLflow, Weights & Biases

Cloud & Infrastructure: AWS (S3, EC2, SQS, Rekognition), Azure AI Services, Google Cloud Platform, Docker, Kubernetes, Terraform, FastAPI, Streamlit

Data & Backend: PostgreSQL, Redis, Apache Spark, Hadoop/MapReduce, Apache Oozie, Apache Kafka, Databricks

EXPERIENCE

Tech Mahindra Americas - AI Engineer -

Mar 2026 – Present

- Architected a multi-agent “Refill” conversational system on FastAPI and PostgreSQL: stage-aware billing/refill journeys, orchestration across agents, and an MCP-style tool gateway (registered tools, invoke/audit path, conversation audit tables) so tool use stays composable and traceable across workflows.
- Implemented persisted conversation memory and context controls in PostgreSQL (turn history, dialog slots, TTL expiry, PII-aware sanitization, capped turn retention, aggregated LLM token usage) to keep chained multi-step calls bounded and cheaper to prompt on follow-up turns.
- Built intent-aware routing for the assistant (billing-focused intents such as balance, refill, plan/usage, bill-change explain, etc.) with configurable routing modes (heuristic vs LLM-first / hybrid), planner integration, and fallbacks when the classifier is unavailable - so the right path is chosen without always paying for the heaviest model call.
- Designed model-agnostic engineering and deployment adapters (prompt normalization, vendor tool-schema mapping, validation, auth/cost/SLA layers, multi-provider examples and demo UI) to reduce provider lock-in and support model swaps and A/B-style routing across the agent stack.
- Operated local and private inference paths using an external NVIDIA GPU setup (Spark / GTX-class hardware) to run large models on-device, with internal hosting for sensitive company workloads so proprietary prompts and data stay off public APIs where policy requires it.

Tawin Solutions LLC (USA) - AI/ML Engineer -

Sept 2025 – Feb 2026

- Designed a personalized restaurant recommendation engine leveraging collaborative and content-based filtering techniques, incorporating user preference history, geolocation signals, and contextual features to improve suggestion relevance.
- Integrated Azure AI services with a structured logging pipeline to capture real-time user interactions and nearby venue data, enabling continuous feedback loops for model retraining and preference drift detection.
- Worked on feature engineering from behavioral data - translating raw choice history into embeddings and structured signals used downstream by the recommendation model.

Tech Mahindra, Makers Lab - AI/ML Intern -

Oct 2023 – Dec 2023

- Contributed to Project Indus, India's first multilingual LLM targeting native Indian languages - involved in the data engineering pipeline (collection, cleaning, and formatting of low-resource language corpora) and participated in transformer model architecture work during pre-training.
- Built a real-time multilingual transcription pipeline for German-to-English meeting audio using Python-based ASR and NLP tooling, focusing on low-latency processing and accurate handling of domain-specific vocabulary.

LTI Mindtree - Cloud Engineering Intern -

Feb 2023 – Apr 2023

- Gained practical experience in cloud infrastructure provisioning across AWS (EC2, S3, IAM, VPC), Azure, and GCP using Terraform writing modular IaC configurations to automate environment setup and ensure reproducibility across deployment targets.

PROJECTS

Intelligent Research Copilot (RAG + NLP Automation)

Built an end-to-end research automation platform that ingests web/doc sources, performs semantic retrieval and abstractive summarization, and serves insights through Streamlit and FastAPI interfaces. Designed a modular AI pipeline for document understanding, retrieval-augmented analysis, and scalable researcher workflows.

City-Scale Traffic Forecasting with Spatiotemporal ML

Developed a spatiotemporal forecasting system for traffic speed prediction using time-series modeling approaches (LSTM/ARIMA) on sensor data. Implemented preprocessing, training, and evaluation workflows to improve short-horizon forecasting reliability for intelligent transportation use cases.

Real-Time Ride Matching Engine (Distributed Systems)

Engineered a low-latency rider-driver matching backend using Redis geospatial indexing, PostgreSQL state management, and WebSocket-based real-time updates. Prototyped distributed coordination and event-driven assignment flows to model production-grade mobility matching behavior.

Event-Driven Vision Pipeline on AWS

Built a cloud-native image/text detection pipeline integrating Amazon S3, Rekognition, and SQS for asynchronous large-batch processing. Implemented service orchestration and confidence-based filtering to support scalable computer-vision inference workflows.

Distributed Flight Analytics with Hadoop + Oozie

Implemented large-scale flight performance analytics using Hadoop MapReduce and orchestrated multi-stage jobs with Apache Oozie. Delivered batch data pipelines for delay/on-time trend analysis across high-volume aviation datasets.

Containerized Spark ML Pipeline for Wine Quality Prediction

Designed a Dockerized PySpark pipeline for model training and inference with cloud-compatible storage integration. Built reproducible batch ML workflows to process tabular data at scale and operationalize experimentation in distributed environments.

AI-Powered Audio Intelligence and Source Separation

Developed an audio intelligence application for music analysis and source separation using Python-based ML/audio tooling and interactive visual analytics. Combined signal processing with deep-learning-based separation workflows to extract instrument-level insights from complex audio tracks.

CERTIFICATIONS

- Azure AI Engineer Associate - Microsoft AI-102
- Azure Data Scientist Associate - Microsoft DP-100
- Databricks Generative AI Fundamentals - Databricks